

ESQ Curriculum Map

This resource includes:

- Learning components assessed in each Milestone
 - *Note: Standards marked with an asterisk (*) indicate that one or more learning components have a restriction*
- Standards, curriculum-specific focus lessons and common misconceptions related to each learning component

How to Use

1. Locate each learning component that appears on your Prepare for Instruction report.
2. Identify the focus lesson(s) for predicted strengths and predicted challenges.
3. Learn about common misconceptions and strategies in the Instructional Moves modules.
4. Plan instruction for each focus lesson and consider...
 - a. Continuing with regular instruction for predicted strengths
 - b. Applying strategies from the Instructional Moves modules to address misconceptions for predicted challenges

[Grade 2 Learning Component Map](#)

[Grade 3 Learning Component Map](#)

[Grade 4 Learning Component Map](#)

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[Grade 7 Learning Component Map](#)

[Grade 8 Learning Component Map](#)

Grade 2 Learning Component Map

Milestone 1	Assess after: Module 1, Topic D
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
2.MD.A.1	Use an appropriate tool to measure the length, in standard units, of an object	2.1.6	View
	Select an appropriate tool to measure the length, in standard units, of an object	2.1.8	View
2.MD.A.2	Describe how two measurements for the same object relate to the size of the units chosen	2.1.10	View
2.MD.A.4	Measure to determine how much longer one object is than another	2.1.14	View

Milestone 2	Assess after: Module 1, Topic I
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
2.NBT.A.1	Given the number of hundreds, tens, and/or ones, write a three- digit number	2.1.25	View
	Decompose a two-digit number into its tens and ones	2.1.25	View
	Decompose a three-digit number into its hundreds, tens, and ones	2.1.25	View
2.NBT.A.4	Compare two three-digit numbers by comparing the digits in each place value	2.1.25 and 2.1.36	View
2.NBT.A.3	Read and write numbers to 1,000 by moving between base-ten numerals (standard form) and expanded form	2.1.26	View

Milestone 2	Assess after: Module 1, Topic I
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
	Read and write numbers to 1,000 by moving between number names (word form) and expanded form	2.1.27	View

Milestone 3	Assess after: Module 2, Topic B
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
2.NBT.B.5	Fluently add within 100 using strategies based on place value, properties of operations, and/or the relationship between addition and subtraction	2.2.3 and 2.2.5	View
2.OA.A.1	Use addition within 100 to solve one-step word problems involving situations of adding to, taking from, putting together, taking apart, and comparing, with unknowns in all positions	2.2.7	View
2.NBT.B.7*	Add within 1,000 using concrete models or drawings and strategies based on place value, properties of operations, and/or the relationship between addition and subtraction (Restricted to addition within 200)	2.2.12	View

Milestone 4	Assess after: Module 2, Topic D
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
2.OA.A.1	Use subtraction within 100 to solve one-step word problems involving situations of adding to, taking from, putting together, taking apart, and comparing, with unknowns in all positions	2.2.13, 2.2.19, and 2.2.26	View

Milestone 4	Assess after: Module 2, Topic D
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
2.NBT.B.5	Fluently subtract within 100 using strategies based on place value, properties of operations, and/or the relationship between addition and subtraction	2.2.15, and 2.2.17	View
2.NBT.B.7*	Subtract within 1,000 using concrete models or drawings and strategies based on place value, properties of operations, and/or the relationship between addition and subtraction (Restricted to subtraction within 200)	2.2.25	View

Milestone 5	Assess after: Module 3, Topic C
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
2.G.A.1	Identify quadrilaterals	2.3.2	View
	Identify hexagons	2.3.2	View
2.G.A.3	Partition objects, including circles and rectangles, into three equal shares and describe the shares using "thirds"	2.3.11	View
	Describe a whole based on its number of parts, including two halves, three thirds, and four fourths	2.3.12	View

Milestone 6	Assess after: Module 4, Topic E
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
2.NBT.B.7	Add or subtract hundreds and hundreds, tens and tens, ones and ones	2.4.7	View
	Add within 1,000 using concrete models or drawings and strategies based on place value,	2.4.8 and 2.4.9	View

Milestone 6		Assess after: Module 4, Topic E	
Standards	Key Learning Components	Focus Lessons	Instructional Moves
	properties of operations, and/or the relationship between addition and subtraction		
	Subtract within 1,000 using concrete models or drawings and strategies based on place value, properties of operations, and/or the relationship between addition and subtraction	2.4.16 and 2.4.17	View

Milestone 7		Assess after: Module 5, Topic C	
Standards	Key Learning Components	Focus Lessons	Instructional Moves
2.MD.B.6	Represent whole numbers as lengths on a number line diagram	2.5.12	View
	Represent whole-number differences on a number line diagram	2.5.12	View
	Represent whole-number sums on a number line diagram	2.5.12	View
2.MD.B.5	Use equations, with a symbol for the unknown number, to represent addition within 100 in word problems involving length		View
	Use subtraction within 100 to solve word problems involving lengths that are given in the same units		View

Milestone 8		Assess after: Module 6, Topic D	
Standards	Key Learning Components	Focus Lessons	Instructional Moves
2.OA.C.4	Write an equation to express a total as a sum of equal addends	2.6.1	View

Milestone 8

Assess after: Module 6, Topic D

Standards	Key Learning Components	Focus Lessons	Instructional Moves
	Determine the total number of objects arranged in rectangular arrays with up to 5 rows and up to 5 columns	2.6.5	View
2.OA.C.3	Determine whether a group of objects (up to 20) has an odd or even number of members	2.6.15	View
	Group objects into pairs to determine if the group of objects (up to 20) has an odd or even number of members	2.6.15	View

Grade 3 Learning Component Map

Milestone 1	Assess after: Module 1, Topic D
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
3.OA.A.1	Model a multiplication situation, including by writing a multiplication expression	3.1.2	View
	Identify a multiplication situation based on a given multiplication expression	3.1.2	View
3.OA.A.2	Identify a division situation based on a given division expression	3.1.9	View
	Interpret whole number quotients as the number of objects in each group when a set number of objects is partitioned into equal groups	3.1.9 and 3.1.17	View
3.OA.B.5	Apply the commutative property as a strategy to multiply	3.1.10 and 3.1.13	View
	Apply the distributive property as a strategy to multiply	3.1.12 and 3.1.14	View

Milestone 2	Assess after: Module 2, Topic B
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
3.MD.A.2	Use subtraction to solve one-step word problems involving masses or volumes that are given in the same units	3.2.3 and 3.2.6	View
	Use multiplication to solve one-step word problems involving masses or volumes that are given in the same units	3.2.3 and 3.2.6	View
3.NBT.A.1	Round whole numbers to the nearest 10	3.2.10	View

Milestone 2		Assess after: Module 2, Topic B	
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
	Round whole numbers to the nearest 100	3.2.11	View

Milestone 3		Assess after: Module 2, Topic D	
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
3.NBT.A.2	Fluently add within 1,000 using strategies and algorithms based on place value, properties of operations, and/or the relationship between addition and subtraction	3.2.14 and 3.2.20	View
	Fluently subtract within 1,000 using strategies and algorithms based on place value, properties of operations, and/or the relationship between addition and subtraction	3.2.14 and 3.2.22	View

Milestone 4		Assess after: Module 3, Topic D	
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
3.OA.B.5	Apply the distributive property as a strategy to multiply	3.3.5	View
	Apply the associative property as a strategy to multiply	3.3.9	View
3.OA.C.7	Fluently divide within 100, using strategies such as the relationship between multiplication and division or properties of operations	3.3.17	View
	Fluently multiply within 100, using strategies such as the relationship between multiplication and division or properties of operations	3.3.24	View

Milestone 4	Assess after: Module 3, Topic D
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
3.OA.D.8	Represent a two-step word problem using equations with a letter standing for the unknown quantity	3.3.19 and 3.3.22	View
	Use addition and subtraction within 1,000 and multiplication and division within 100 to solve two step real- world problems	3.3.19 and 3.3.22	View

Milestone 5	Assess after: Module 4, Topic D
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
3.MD.C.6	Measure the area of a rectangle by counting unit squares	3.4.4	View
3.MD.C.7.B	Use multiplication to find the area of a rectangle with whole number side lengths in the context of solving mathematical problems	3.4.9	View
	Use multiplication to find the area of a rectangle with whole number side lengths in the context of solving real-world problems	3.4.13	View
3.MD.C.7.D	Find the area of rectilinear figures by decomposing them into non-overlapping parts and finding the area of each part	3.4.14 and 3.4.15	View

Milestone 6	Assess after: Module 5, Topic B
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
3.G.A.2	Partition shapes into parts with equal areas (2, 3, 4, 6, or 8 equal parts)	3.5.1	View

Milestone 6

Assess after: Module 5, Topic B

Standards	Key Learning Components	Focus Lessons	Instructional Moves
	Express the area of each equal part of a shape as a unit fraction of the whole (with denominators of 2, 3, 4, 6, and 8)	3.5.5	View
3.NF.A.1	Identify a fraction $\frac{1}{b}$ as the quantity formed by 1 part when a whole is partitioned into b equal parts (where b is 2, 3, 4, 6, or 8)	3.5.5	View
	Identify a fraction $\frac{a}{b}$ as the quantity formed by a parts of size $\frac{1}{b}$ (where b is 2, 3, 4, 6, or 8)	3.5.6	View

Milestone 7

Assess after: Module 5, Topic E

Standards	Key Learning Components	Focus Lessons	Instructional Moves
3.NF.A.2.A	Recognize that when the interval 0 to 1 is partitioned into b equal parts, each part has size $\frac{1}{b}$	3.5.11	View
3.NF.A.3.B	Represent a fraction $\frac{a}{b}$ with a model, including a number line (where b is 2, 3, 4, 6, or 8)	3.5.12	View
	Recognize simple equivalent fractions (with denominators 2, 3, 4, 6, and 8)	3.5.13	View
	Generate simple equivalent fractions (with denominators 2, 3, 4, 6, and 8)	3.5.13 and 3.5.24	View
3.NF.A.3.D	Compare two fractions with the same numerator by reasoning about their size	3.5.19	View

Milestone 8

Assess after: Module 6, Topic C

Standards	Key Learning Components	Focus Lessons	Instructional Moves
3.MD.A.1	Tell and write time to the nearest minute	3.6.2	View
	Measure time intervals in minutes (within 60 minutes), including finding elapsed time	3.6.5	View
3.G.A.1	Understand that shapes in different categories may share attributes	3.6.8	View
3.MD.D.8	Solve mathematical problems involving the perimeters of polygons	3.6.15 and 3.6.18	View
	Differentiate between perimeter and area and recognize that figures can have the same perimeter and different areas	3.6.16 and 3.6.17	View
	Solve real-world problems involving finding the missing side length of a polygon when the perimeter is given	3.6.18	View

Grade 4 Learning Component Map

Milestone 1	Assess after: Module 1, Topic C
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
4.NBT.A.1	Apply concepts of place value to recognize quotients and products in multiplication and division problems involving whole numbers and a factor of 10	4.1.6	View
	Recognize that in a multi-digit whole number, a digit in one place represents ten times what it represents in the place to its right	4.1.6	View
4.NBT.A.2	Read and write multi-digit numbers up to 1,000,000 by moving between base-ten numerals (standard form) and expanded form	4.1.7	View
	Compare two multi-digit numbers	4.1.9	View
4.NBT.A.3	Round multi-digit whole numbers to any place within 1,000,000	4.1.14	View

Milestone 2	Assess after: Module 2, Topic B
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
4.NBT.B.4	Fluently add multi-digit whole numbers (sums to 1,000,000) using the standard algorithm	4.1.16	View
	Fluently subtract multi-digit whole numbers (minuend to 1,000,000) using the standard algorithm	4.1.18	View
4.NBT.B.5*	Multiply a three-digit number by a one-digit number using strategies based on place value or properties of operations	4.2.5 and 4.2.6	View

Milestone 2	Assess after: Module 2, Topic B
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
	(Restricted to multiplication of two-digit numbers by one-digit numbers)		

Milestone 3	Assess after: Module 3, Topic B
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
4.OA.B.4	Determine whether a whole number within 100 is prime or composite	4.2.21	View
	Find all factor pairs of a whole number within 100	4.2.21	View
	Determine whether a whole number within 100 is a multiple of a given one-digit number	4.2.23	View
4.NBT.B.6	Find whole-number quotients (no remainders) of four-digit dividends and one-digit divisors using strategies based on place value, the properties of operations, and/or the relationship between multiplication and division.	4.3.5	View

Milestone 4	Assess after: Module 3, Topic F
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
4.NBT.B.5	Use models to explain the multiplication of three-digit numbers by one-digit numbers	4.3.9	View
	Multiply a four-digit number by a one-digit number using strategies based on place value or properties of operations	4.3.10	View

Milestone 4

Assess after: Module 3, Topic F

Standards	Key Learning Components	Focus Lessons	Instructional Moves
	Multiply a two-digit number by a two-digit number using strategies based on place value or properties of operations	4.3.14	View
4.NBT.B.6	Solve multi-step real-world problems using the four operations in situations where a remainder must be interpreted	4.3.23	View
4.OA.A.3	Solve multi-step real-world problems using the four operations in situations where a remainder must be interpreted	4.3.23	View
	Represent multi-step real-world problems using the four operations with an equation with a letter standing for the unknown	4.3.24	View

Milestone 5

Assess after: Module 4, Topic C

Standards	Key Learning Components	Focus Lessons	Instructional Moves
4.NF.B.3.B	Explain why a fraction a/b is equivalent to a fraction $(n \times a)/(n \times b)$ using visual fraction models	4.4.3	View
4.NF.A.1	Explain why a fraction a/b is equivalent to a fraction $(n \times a)/(n \times b)$ using visual fraction models	4.4.8	View
	Generate equivalent fractions using the principle $a/b = (n \times a)/(n \times b)$	4.4.8	View
4.NF.A.2	Compare two fractions with different numerators and different denominators	4.4.16	View

Milestone 6	Assess after: Module 4, Topic E
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
4.NF.B.3.D	Solve word problems involving addition of fractions referring to the same whole and having like denominators	4.4.21	View
	Solve word problems involving subtraction of fractions referring to the same whole and having like denominators	4.4.21	View
4.NF.B.3.C	Add mixed numbers with like denominators	4.4.23	View
	Subtract mixed numbers with like denominators	4.4.25 and 4.4.26	View

Milestone 7	Assess after: Module 5, Topic D
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
4.NF.C.6	Use decimal notation for fractions with a denominator of 100	4.5.5	View
	Locate decimals to the tenths or hundredths place on a number line	4.5.5	View
4.NF.C.7	Compare two decimals to the hundredths place	4.5.11	View
4.NF.C.5	Add two fractions with respective denominators 10 and 100	4.5.12	View

Milestone 8	Assess after: Module 6, Topic D
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
4.G.A.1	Draw rays and identify them in two-dimensional figures	4.6.1	View

Milestone 8

Assess after: Module 6, Topic D

Standards	Key Learning Components	Focus Lessons	Instructional Moves
	Draw parallel and perpendicular lines and identify them in two-dimensional figures	4.6.4 and 4.6.5	View
4.MD.C.7	Solve addition problems to find unknown angles on a diagram in real-world problems	4.6.14 and 4.6.15	View
	Represent addition problems to find unknown angles on a diagram using an equation with a symbol for the unknown angle measure	4.6.14 and 4.6.15	View
4.G.A.2	Identify right triangles	4.6.18	View
	Classify two-dimensional figures based on the presence or absence of perpendicular or parallel lines	4.6.20	View

Grade 5 Learning Component Map

Milestone 1		Assess after: Module 1, Topic C	
Standards	Key Learning Components	Focus Lessons	Instructional Moves
5.NBT.B.5	Fluently multiply 2-digit by up to 3-digit whole numbers using the standard algorithm	5.1.9	View
	Fluently multiply 3-digit by 3-digit whole numbers using the standard algorithm	5.1.10	View
	Fluently multiply 3-digit by 4-digit whole numbers using the standard algorithm	5.1.10	View
5.NBT.B.6	Find whole number quotients of whole numbers with three-digit dividends and two-digit divisors using strategies based on place value, the properties of operations, and/or the relationship between multiplication and division	5.1.14, 5.1.15	View
	Find whole number quotients of whole numbers with four-digit dividends and two-digit divisors using strategies based on place value, the properties of operations, and/or the relationship between multiplication and division	5.1.16	View
Milestone 2		Assess after: Module 2, Topic C	
Standards	Key Learning Components	Focus Lessons	Instructional Moves
5.NF.A.1	Add fractions with different denominators by using equivalent fractions	5.2.7 and 5.2.9	View
	Subtract fractions with different denominators by using equivalent fractions	5.2.7 and 5.2.9	View
	Subtract mixed numbers with different denominators by using equivalent fractions	5.2.13 and 5.2.14	View

Milestone 2

Assess after: Module 2, Topic C

Standards	Key Learning Components	Focus Lessons	Instructional Moves
5.NF.A.2	Solve word problems involving addition of fractions referring to the same whole, including with different denominators	5.2.11	View
	Solve word problems involving subtraction of fractions referring to the same whole, including with different denominators	5.2.14	View

Milestone 3

Assess after: Module 3, Topic C

Standards	Key Learning Components	Focus Lessons	Instructional Moves
5.NF.B.4.A	Interpret the product $(a/b) \times q$ as the result of a sequence of operations $a \times q \div b$, where q represents either a whole number or a fraction	5.3.3	View
5.NF.B.7.A	Compute quotients of a unit fraction and a non-zero whole number	5.3.14	View
5.NF.B.7.B	Compute the quotient of a whole number dividend and a unit fraction divisor	5.3.15	View
5.NF.B.6	Solve a real-world problem involving multiplication of fractions and mixed numbers	5.3.17	View
	Use equations to represent and solve a real-world fraction or mixed-number multiplication problem	5.3.17	View
5.NF.B.7.C	Solve real-world problems involving division of whole numbers by unit fractions	5.3.17	View

Milestone 4

Assess after: Module 4, Topic B

Standards	Key Learning Components	Focus Lessons	Instructional Moves
5.NBT.A.3.A	Read and write decimals to thousandths by moving between base-ten numerals (standard form) and number names (word form)	5.4.2	View
	Read and write decimals to the thousandths by moving between base-ten numerals (standard form) and expanded form	5.4.3	View
5.NBT.A.3.B	Compare two decimals to the thousandths	5.4.6	View
5.NBT.B.7	Add decimals to the hundredths using concrete models or drawings and strategies based on place value, properties of operations, and/or the relationship between addition and subtraction	5.4.9 and 5.4.10	View
	Subtract decimals to the hundredths using concrete models or drawings and strategies based on place value, properties of operations, and/or the relationship between addition and subtraction	5.4.11 and 5.4.12	View

Milestone 5

Assess after: Module 4, Topic E

Standards	Key Learning Components	Focus Lessons	Instructional Moves
5.NBT.B.7	Multiply decimals to the hundredths using concrete models or drawings and strategies based on place value and/or properties of operations	5.4.17 and 5.4.19	View
	Divide decimals to the hundredths using concrete models or drawings and strategies based on place value and/or properties of operations	5.4.24 and 5.4.25	View
5.MD.A.1	Convert a measurement of length, mass, or volume from a smaller unit to a larger unit within the same system of measurement	5.4.27 and 5.4.28	View

Milestone 5		Assess after: Module 4, Topic E	
Standards	Key Learning Components	Focus Lessons	Instructional Moves
	Convert a measurement of length, mass, or volume from a larger unit to a smaller unit within the same system of measurement	5.4.27 and 5.4.28	View

Milestone 6		Assess after: Module 5, Topic B	
Standards	Key Learning Components	Focus Lessons	Instructional Moves
5.G.B.4	Classify quadrilaterals in a hierarchy based on properties	5.5.7	View
5.NF.B.4.B	Find the area of a rectangle with fractional side lengths by tiling it with unit squares of the appropriate unit fraction side lengths	5.5.10	View
	Multiply fractional side lengths to find areas of rectangles	5.5.11	View
5.NF.B.4.b	Represent fraction products as rectangular areas	5.5.10	View

Milestone 7		Assess after: Module 5, Topic D	
Standards	Key Learning Components	Focus Lessons	Instructional Moves
5.MD.C.5.A	Find the volume of a right rectangular prism with whole-number side lengths by packing it with unit cubes	5.5.17	View
	Represent threefold whole-number products as volume	5.5.23	View

Milestone 7

Assess after: Module 5, Topic D

Standards	Key Learning Components	Focus Lessons	Instructional Moves
5.MD.C.5.B	Apply the formula $V = b \times h$ to find volumes of right rectangular prisms with whole number edge lengths in the context of solving real-world problems	5.5.22	View
	Apply the formula $V = l \times w \times h$ to find volumes of right rectangular prisms with whole number edge lengths in the context of solving mathematical problems	5.5.23	View
5.MD.C.5.C	Find volumes of solid figures composed of two non-overlapping right rectangular prisms by adding the volumes of the non-overlapping parts	5.5.25	View
	Solve real-world problems by finding the volumes of solid figures composed of two non-overlapping right rectangular prisms by adding the volumes of the non-overlapping parts	5.5.26	View

Milestone 8

Assess after: Module 6, Topic D

Standards	Key Learning Components	Focus Lessons	Instructional Moves
5.G.A.1	Interpret the x-coordinate of an ordered pair as the distance to travel along the x-axis from the origin and the y-coordinate of an ordered pair as the distance to travel along the y-axis from the origin	5.6.2	View
5.OA.B.3	Form ordered pairs consisting of corresponding terms from two patterns and graph the ordered pairs on a coordinate plane	5.6.7	View
	Identify apparent relationships between corresponding terms	5.6.9	View

Standards	Key Learning Components	Focus Lessons	Instructional Moves
5.G.A.2	Interpret coordinate values of points in the context of the situation	5.6.16	View
	Represent mathematical problems by graphing points in the first quadrant of the coordinate plane	5.6.16	View

Grade 6 Learning Component Map

Milestone 1

Assess after: Module 1, Topic C

Standards	Key Learning Components	Focus Lessons	Instructional Moves
6.RP.A.1	Use ratio language to describe a ratio relationship between two quantities	6.1.2	View
	Write a ratio to represent the relationship between two quantities	6.1.2 and 6.1.3	View
6.RP.A.3.a	Find missing values in ratio tables	6.1.8 and 6.1.9	View
	Use tables to compare ratios	6.1.13 and 6.1.14	View

Milestone 2

Assess after: Module 1, Topic E

Standards	Key Learning Components	Focus Lessons	Instructional Moves
6.RP.A.3.b	Solve unit rate problems including those involving unit pricing and constant speed	6.1.20	View
6.RP.A.3.c	Find a percent of a quantity as a rate per 100	6.1.23	View
	Solve problems involving finding the whole when given a part and the percent	6.1.24 and 6.1.25	View

Milestone 3

Assess after: Module 2, Topic F

Standards	Key Learning Components	Focus Lessons	Instructional Moves
6.NS.B.3	Fluently add multi-digit decimals using the standard algorithm	6.2.13	View
	Fluently subtract multi-digit decimals using the standard algorithm	6.2.13	View

Milestone 3

Assess after: Module 2, Topic F

Standards	Key Learning Components	Focus Lessons	Instructional Moves
	Fluently multiply multi-digit decimals using the standard algorithm	6.2.15	View
	Fluently divide multi-digit decimals using the standard algorithm	6.2.21, 6.2.22, and 6.2.23	View
6.NS.B.2	Fluently divide multi-digit whole numbers using the standard algorithm	6.2.18 and 6.2.19	View

Milestone 4

Assess after: Module 3, Topic D

Standards	Key Learning Components	Focus Lessons	Instructional Moves
6.NS.C.6.a	Recognize that the opposite of the opposite of a number is the number itself	6.3.2	View
	Recognize that opposite signs of numbers are located on opposite sides of 0 on the number line	6.3.2	View
6.NS.C.6.c	Find and position integers and other rational numbers on a vertical or horizontal number line diagram	6.3.3	View
	Find and position integers and other rational numbers on a coordinate plane	6.3.11	View
6.NS.C.6.b	Use the signs of numbers in an ordered pair to identify the quadrant of the point	6.3.10	View
	Use the signs of numbers in ordered pairs to identify when two ordered pairs represent reflections across one or both axes	6.3.12	View

Milestone 5

Assess after: Module 4, Topic C

Standards	Key Learning Components	Focus Lessons	Instructional Moves
6.EE.A.2.a	Identify a phrase that describes a given expression with operations, numerical values and with letters standing for numbers	6.4.7	View
	Identify an expression that records operations with numerical values and with letters standing for numbers for a given phrase	6.4.8	View
6.EE.A.2.c	Evaluate expressions at specific values of their variables in mathematical contexts	6.4.8	View
	Evaluate expressions at specific values of their variables in real-world contexts	6.4.12	View

Milestone 6

Assess after: Module 5, Topic A

Standards	Key Learning Components	Focus Lessons	Instructional Moves
6.EE.B.7	Solve mathematical problems by solving equations in the form $px = q$ for cases in which p , q , and x are all nonnegative rational numbers	6.4.20	View
	Solve real-world problems by solving equations in the form $x + p = q$ for cases in which p , q , and x are all nonnegative rational numbers	6.4.21	View
	Represent real-world problems by writing equations in the form $px = q$ for cases in which p , q , and x are all nonnegative rational numbers	6.4.21	View
6.EE.C.9	Represent a real-world situation by writing an equation that expresses the dependent variable in terms of the independent variable	6.4.22 and 6.4.24	View

Milestone 6		Assess after: Module 5, Topic A	
Standards	Key Learning Components	Focus Lessons	Instructional Moves
	Analyze the relationship between dependent and independent variables using graphs and tables and relate to an equation	6.4.23 and 6.4.24	View
	Model the relationship between independent and dependent variables to solve real-world problems	6.4.25	View
Milestone 7		Assess after: Module 5, Topic D	
Standards	Key Learning Components	Focus Lessons	Instructional Moves
6.G.A.1	Find area of quadrilaterals by composing into rectangles or decomposing into triangles and other shapes and use to solve mathematical problems	6.5.6 and 6.5.7	View
	Find area of polygons by composing into rectangles or decomposing into triangles and other shapes and use to solve real-world problems	6.5.8	View
6.G.A.4	Represent three-dimensional figures using nets made up of rectangles and triangles	6.5.10	View
	Use nets to find the surface area of three-dimensional figures	6.5.11	View
Milestone 8		Assess after: Module 6, Topic D	
Standards	Key Learning Components	Focus Lessons	Instructional Moves
6.SPA.2	Recognize a data distribution that is symmetric, skewed left, or skewed right	6.6.3	View

Milestone 8

Assess after: Module 6, Topic D

Standards	Key Learning Components	Focus Lessons	Instructional Moves
6.SP.B.5.c	Summarize numerical data sets in relation to their context by calculating quantitative measures of center (median and/or mean)	6.6.7, 6.6.8, and 6.6.12	View
	Summarize numerical data sets in relation to their context by calculating quantitative measures of variability (interquartile range and/or mean absolute deviation)	6.6.10 and 6.6.13	View
6.SP.B.5.d	Summarize numerical data sets in relation to their context by relating the choice of measures of center to the shape of the data distribution and the context in which the data were gathered	6.6.20	View

Grade 7 Learning Component Map

Milestone 1	Assess after: Module 1, Topic C
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
7.RP.A.2c	Represent proportional relationships with equations	7.1.2 and 7.1.3	View
7.RP.A.2.B	Identify or determine the constant of proportionality (unit rate) in tables of proportional relationships	7.1.3	View
	Identify or determine the constant of proportionality (unit rate) in graphs of proportional relationships	7.1.5	View
7.G.A.1	Compute actual lengths from a scale drawing (using proportional reasoning)	7.1.17	View
	Compute actual areas from a scale drawing (using proportional reasoning)	7.1.18	View

Milestone 2	Assess after: Module 2, Topic B
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
7.NS.A.1.B	Interpret $p + q$ as the number located a distance $ q $ from p in the positive or negative direction depending on whether q is positive or negative	7.2.1	View
	Interpret sums of rational numbers (fractions, decimals, and integers) by describing them with real-world contexts	7.2.3	View
7.NS.A.1.D	Use properties of operations to add rational numbers	7.2.6	View

Milestone 2	Assess after: Module 2, Topic B
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
	Use properties of operations to subtract rational numbers	7.2.10 and 7.2.11	View
7.NS.A.1.C	Understand $p - q$ is equivalent to $p + (-q)$	7.2.8 and 7.2.9	View

Milestone 3	Assess after: Module 3, Topic A
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
7.NS.A.2.C	Apply properties of operations as strategies to multiply rational numbers	7.2.15	View
	Apply properties of operations as strategies to divide rational numbers	7.2.18, 7.2.19, and 7.2.22	View
7.EE.A.1	Expand linear expressions with rational coefficients	7.3.2	View
	Add linear expressions with rational coefficients	7.3.4	View
	Factor linear expressions with rational coefficients	7.3.5	View

Milestone 4	Assess after: Module 3, Topic D
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
7.EE.B.4.A	Solve equations of the forms $px + q = r$ and $p(x + q) = r$ fluently	7.3.13, 7.3.14, and 7.3.15	View
	Write multi-step equations to represent real-world problems, including using rational numbers	7.3.16	View
	Graph the solution set of a multi-step inequality	7.3.18	View

Milestone 4	Assess after: Module 3, Topic D
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
7.EE.B.4.B	Write multi-step inequalities to represent real-world problems, including using rational numbers	7.3.22	View

Milestone 5	Assess after: Module 4, Topic C
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
7.G.A.2	Know and understand conditions that determine a unique triangle, more than one triangle, or no triangle	7.4.6	View
7.G.B.4	Find the area of a circle to solve problems, including by using the formula for area of a circle	7.4.12	View
	Find the circumference of a circle to solve problems, including by using the formula for circumference of a circle	7.4.13	View

Milestone 6	Assess after: Module 4, Topic E
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
7.G.B.6	Determine the area of two-dimensional figures composed of various polygons in real-world problems	7.4.16	View
	Determine the surface area of three-dimensional prisms in mathematical problems	7.4.17	View
	Determine the volume of three-dimensional objects composed of cubes and right prisms in real-world problems	7.4.26	View

Milestone 6

Assess after: Module 4, Topic E

Standards	Key Learning Components	Focus Lessons	Instructional Moves
7.G.A.3	Describe and identify the two-dimensional figures that result from slicing a right rectangular prism or pyramid by a plane perpendicular to the base or a plane parallel to the base	7.4.22	View

Milestone 7

Assess after: Module 5, Topic E

Standards	Key Learning Components	Focus Lessons	Instructional Moves
7.RP.A.3	Solve multi-step ratio problems, including by using proportional relationships	7.5.6 and 7.5.17	View
	Solve multi-step percent problems, including by using proportional relationships	7.5.11 and 7.5.12	View
7.EE.A.2	Rewrite an expression in different forms in a problem context	7.5.11	View
	Interpret how quantities in an expression are related	7.5.16	View

Milestone 8

Assess after: Module 6, Topic D

Standards	Key Learning Components	Focus Lessons	Instructional Moves
7.SP.C.8	Find probabilities of compound independent events	7.6.5	View
7.SP.A.1	Understand that generalizations about a population from a sample are valid only if the sample is representative of that population	7.6.11	View

Milestone 8

Assess after: Module 6, Topic D

Standards	Key Learning Components	Focus Lessons	Instructional Moves
7.SP.A.2	Use data from a random sample to draw inferences about a population with an unknown characteristic of interest	7.6.14	View
7.SP.B.3	Measure the difference between the centers of two numerical data distributions by expressing it as a multiple of a measure of variability	7.6.18	View
7.SP.B.4	Draw informal comparative inferences about numerical data from random samples of two populations, including by using measures of center and variability	7.6.19	View

Grade 8 Learning Component Map

Milestone 1	Assess after: Module 1, Topic C
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
8.EE.A.4	Add, subtract, multiply, and divide numbers written in scientific notation	8.1.4 and 8.1.12	View
8.EE.A.1	Use exponent properties to generate equivalent numerical expressions with integer exponents	8.1.9	View

Milestone 2	Assess after: Module 1, Topic E
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
8.EE.A.2	Recognize and evaluate the roots of perfect squares and cubes	8.1.16	View
	Represent the solutions of simple quadratic and cubic expressions by using square and cube roots	8.1.17	View
8.NS.A.1	Know that numbers that are not rational are called irrational, and identify irrational numbers	8.1.22	View
	Understand a rational number to be any number that is the value of a ratio of two integers	8.1.22	View
	Convert a decimal expansion which repeats eventually into a rational number	8.1.22	View

Milestone 3	Assess after: Module 2, Topic D
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
8.G.A.2	Describe a sequence of transformations that exhibits congruence between two given congruent figures	8.2.10	View
8.G.B.7	Use the Pythagorean Theorem to solve for an unknown side length in right triangles in real-world problems in two and three dimensions	8.2.19	View
	Use the Pythagorean Theorem to solve for an unknown side length in right triangles in mathematical problems in two and three dimensions	8.2.21	View

Milestone 4	Assess after: Module 3, Topic D
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Standards	Key Learning Components	Focus Lessons	Instructional Moves
8.G.A.3	Describe the effect of dilation on two-dimensional figures in the coordinate plane, including the use of coordinates and algebraic representation	8.3.7 and 8.3.8	View
8.G.A.4	Understand that a two-dimensional figure is similar to another if the second can be obtained from the first by a sequence of rotations, reflections, translations, and dilations	8.3.11	View
	Given two similar two-dimensional figures, describe a sequence that exhibits the similarity between them	8.3.11	View
8.G.A.5	Establish and use the angle-angle criterion for similarity of triangles	8.3.12	View

Milestone 5

Assess after: Module 4, Topic D

Standards	Key Learning Components	Focus Lessons	Instructional Moves
8.EE.C.7.B	Solve a linear equation with rational coefficients	8.4.3	View
8.EE.C.7.A	Isolate the single variable in a linear equation and interpret the result to determine its number of solutions	8.4.7 and 8.4.8	View
	Generate examples of a linear equation in one variable with one solution, many solutions and no solutions	8.4.7 and 8.4.8	View
8.EE.B.5	Compare two proportional relationships represented in different ways	8.4.15	View
	Interpret the slope of the graph of a proportional relationship as the unit rate	8.4.16	View

Milestone 6

Assess after: Module 5, Topic B

Standards	Key Learning Components	Focus Lessons	Instructional Moves
8.EE.B.6	Use similar triangles to explain why the slope of a line graphed on the coordinate plane is consistent	8.4.16 and 8.4.20	View
	Write the equation $y = mx + b$ for a line graphed not through the origin	8.4.20	View
	Write the equation $y = mx$ for a line graphed through the origin	8.4.20	View
8.EE.C.8.B	Solve simple systems of linear equations by inspection	8.5.3 and 8.5.10	View
	Identify solutions to systems of linear equations visually on graphs	8.5.5	View
	Solve systems of linear equations algebraically	8.5.6 and 8.5.9	View

Milestone 7

Assess after: Module 6, Topic B

Standards	Key Learning Components	Focus Lessons	Instructional Moves
8.F.A.3	Identify the equation $y = mx + b$ as representing a linear function	8.6.3	View
	Identify the graph of a line as representing a linear function	8.6.5	View
	Recognize and give examples of functions that are (or are not) linear	8.6.10	View
8.F.B.4	Determine the initial value of a function from a real-world context presented in a table or graph	8.6.7	View
	Model a linear relationship between two quantities with a function	8.6.7	View
	Interpret the rate of change of a linear function in the context of the situation it models	8.6.7	View

Milestone 8

Assess after: Module 6, Topic E

Standards	Key Learning Components	Focus Lessons	Instructional Moves
8.SPA.1	Interpret scatterplots and recognize patterns of association in the data	8.6.11	View
	Describe the linearity, strength, direction, and clusters or outliers of an association shown in a scatter plot	8.6.12	View
8.SPA.2	Informally fit a straight line to model a data set	8.6.13	View
	Informally assess the fit of a straight line used to model a data set by judging the closeness of the data points to the line	8.6.15	View

Standards	Key Learning Components	Focus Lessons	Instructional Moves
8.SP.A.3	Interpret the slope of a line of best fit	8.6.14	View
	Interpret the intercept of a line of best fit	8.6.14	View